11	**(a)**		$A = 11$ Substitution of $t = 3$ and $V = 13.8$ $B = 7.0$	**B1** **M1** **A1** **[3]**	**1.1** **3.3** **1.1**	from $t = 0$ and $V = 11$ $13.8 = A + B(1 - e^{-0.17\times3})$ Allow 7, 7.01, 7.009, etc	Implied by 7.0086
11	**(b)**		$t = 4$ and $V = 14.453681053\ldots$to 1 or more dp good fit $t = 5$ and $V = 15.008095476..$to 1 or more dp good fit	**B1** **B1** **[2]**	**3.4** **1.1**	allow **SC1** for two correct values with no comment or incorrect comment(s) If B0B0, allow sc M1 for subst with their values.	Allow full marks if more accurate value of B used – 14.458 and 15.013
11	**(c)**		$\frac{dV}{dt}$ = their $7.0 \times 0.17\ e^{-0.85}$ 0.509 ms^{-2}	**M1** **A1** **[2]**	**3.1a** **1.1**	Allow for their $7.0 \times 0.17\ e^{-0.17t}$; condone use of 7.0086 for B	

Question			Answer	Marks	AOs	Guidance	
11	(d)		$t \to \infty$, $V \to 11 + 7$ oe	M1	3.3	Allow ft from (a) for M1M1	
			their 18×3.6	M1	1.1	Allow for any attempt to change to km/hr, accept use of 'their 18'	
			$64.8 > 60$ so the motorist is fined	A1	3.5a	Need to show $64.8 > 60$ for full marks	
				[3]			